Simple and Effective Extrapolation Technique for Neural-Based Microwave Modeling

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***Abstract*—This letter presents a simple and effective technique allowing neural network based microwave models to be used outside their training range. Our technique detects necessity of extrapolation inside each hidden neuron and performs extrapolation using a modified neuron activation function. Compared to existing techniques, the proposed technique is easy to implement and the extrapolation result is equally good or better. Examples of extrapolating neural models of a high electron mobility transistor and a power amplifier are presented.**

***Index Terms*—Extrapolation, microwave modeling, neural networks, neuron.**

I. Introduction

NEURAL networks (NN) have been recognized in microwave CAD area as a useful vehicle to address the computational challenges in modeling, simulation, and optimization [1]. They can be trained to learn EM/physics behaviors from component data. Trained neural networks can be used for high-level circuit and system designs providing faster results than original EM/physics simulations [2]–[6].

In general, a neural network model has good accuracy only if the model inputs are within the region where it is trained. Outside this region, the accuracy of the model deteriorates quickly. This creates limitations of the model in cases where microwave designers explore design solutions in a wider range than the model training range. Studies have been conducted on the extrapolation [7] of neural networks, such as for short-term prediction of financial applications [8], extrapolation detection [9], and microwave modeling [10]. Knowledge-based neural networks [1] use microwave empirical formulas with neural networks to help model extrapolation. This method depends on the availability of empirical or equivalent models. In the absence of such a model, mathematical extrapolation methods can be used, based on the model and its derivative information at the boundary of the neural network training range in a multi-dimensional input space [9], [10]. However, in microwave modeling such as dynamic neural network (DNN) for nonlinear transistor and power amplifier modeling [6], the multi-dimensional training boundary can be irregular or unknown [10]. In this case, existing extrapolation approaches become less effective or complicated.

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In this letter, a simple and effective extrapolation technique for neural-based microwave modeling is proposed. It is formulated for each hidden neuron using reliable one-dimensional extrapolation along the neuron switch direction. The proposed formulation is easy to implement. Simple criteria of identifying directions for extrapolation and which hidden neurons to extrapolate are described.

II. Proposed Neuron-Level Extrapolation for Neural Network Model

The poor performance of neural networks outside training range is due to the switch functions inside hidden neurons [1]. These functions are important for achieving model accuracy within the training range. However, they eventually saturate to constant values as the neural network inputs go far beyond the training range. Consequently, the neural network cannot extrapolate the tendency of the original problem.

This section proposes a novel neuron-level extrapolation technique to modify the neural network responses outside the training range. Instead of relying on the detection of the training boundary of the entire model input space [8]–[10], we exploit the information inside the hidden neurons.

Consider a three-layer perceptron neural network [1] with one input layer, one hidden layer, and one output layer. Let N_x, N_h, and N_y represent the number of input neurons, hidden neurons, and output neurons, respectively. Each hidden neuron of the neural network behaves as a one-dimensional switch along a certain direction in the model input space. For the ith ($i = 1, 2, \ldots, N_h$) hidden neuron, the specific direction is dictated by the neural network weights $[w_{i1} w_{i2} \ldots w_{iNx}]$, where $w_{ij}(j = 1, 2, \ldots, N_x)$ represents the weight parameter connecting the ith hidden neuron and jth input neuron. Let x_j represent the input given to the jth input neuron. Let $\sigma(\gamma_i)$ represent the neuron activation function, e.g., sigmoid function [1], as the one-dimensional switch of the ith hidden neuron. Let γ_i be the one-dimensional variable for the switch function defined as

$$\gamma_i = \sum_{j=1}^{N_x} w_{ij} \cdot x_j. \qquad (1)$$

Since the values of $[w_{i1} w_{i2} \ldots w_{iNx}]$ are different for different hidden neurons i ($i = 1, 2, \ldots, N_h$), various hidden neurons will switch along different directions in the model input space.

Fig. 1 illustrates the proposed neuron-level extrapolation for the ith ($i = 1, 2, \ldots, N_h$) hidden neuron of the neural network. It first evaluates the switch variable γ_i from all training samples. Define N_{Tr} as the total number of training samples and k ($k = 1, 2, \ldots, N_{Tr}$) as the sample index. Let $\boldsymbol{x}^k$ be the model inputs of the kth training sample. For $\boldsymbol{x} = \boldsymbol{x}^k$, we propose to define

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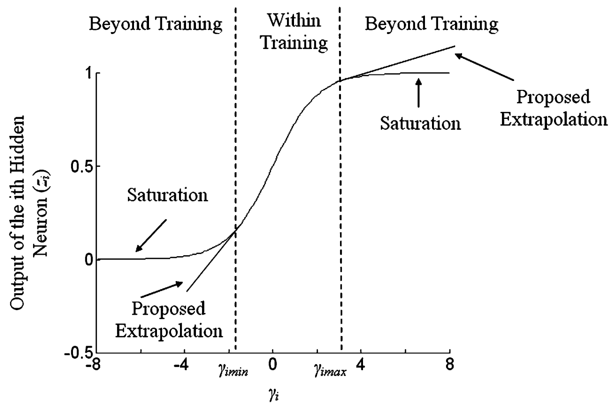

Fig. 1. Illustration of the proposed neuron-level extrapolation technique for a hidden neuron. The saturation region outside training range is replaced by the linear approximation extended from the γ-boundary of the hidden neuron.

the one-dimensional training boundary of the ith hidden neuron, called the γ-boundary, as

$$\gamma_{i\min} = \min_{1 \le k \le N_{T_r}} \left(\sum_{j=1}^{N_x} w_{ij} \cdot x_j^k \right) \tag{2}$$

$$\gamma_{i\max} = \max_{1 \le k \le N_{T_r}} \left(\sum_{j=1}^{N_x} w_{ij} \cdot x_j^k \right) \tag{3}$$

where $\gamma_{i\min}(\gamma_{i\max})$ represents the minimum (maximum) of γ_i.

When neural network inputs $\boldsymbol{x}$ go beyond the training range along a certain direction, only the hidden neurons related to that direction need to be evaluated beyond their γ-boundaries. We identify these hidden neurons by comparing their γ-values w.r.t. their γ-boundaries. For the hidden neuron whose γ-value is outside the corresponding γ-boundary, we perform linear extrapolation using the value and derivative of the neuron activation function at its γ-boundary. For other neurons whose γ-values are inside their γ-boundaries, the standard neuron activation function is used. The modified neuron activation function with extrapolation is

$$z_i = \begin{cases} \sigma(\gamma_{i\min}) + \sigma(\gamma_{i\min}) \cdot (1 - \sigma(\gamma_{i\min})) \cdot (\gamma_i - \gamma_{i\min}), & \text{if } \gamma_{\mathrm{i}} < \gamma_{i\min} \\ \sigma(\gamma_i), & \text{if } \gamma_{i\min} \le \gamma_{\mathrm{i}} \le \gamma_{i\max} \\ \sigma(\gamma_{i\max}) + \sigma(\gamma_{i\max}) \cdot (1 - \sigma(\gamma_{i\max})) \cdot (\gamma_i - \gamma_{i\max}), & \text{if } \gamma_{\mathrm{i}} > \gamma_{i\max} \end{cases} \tag{4}$$

where the term $\sigma \cdot (1 - \sigma)$ is the derivative of the function, i.e., the sigmoid function. Since each hidden neuron switches only along one dimension, i.e., γ_i determined by (1), the extrapolation formula is simply a one-variable extrapolation, with the precise training boundary for the one-variable being easily determined by (2) and (3). If the neural network has only one input, the effective range of the extrapolation is equal to that of the 1st order Taylor expansion at the boundary of the training range. If the neural network has multiple inputs, the proposed technique simplifies the problem into a combination of several one-dimensional 1st order extrapolations along the neuron switch directions of a set of hidden neurons.

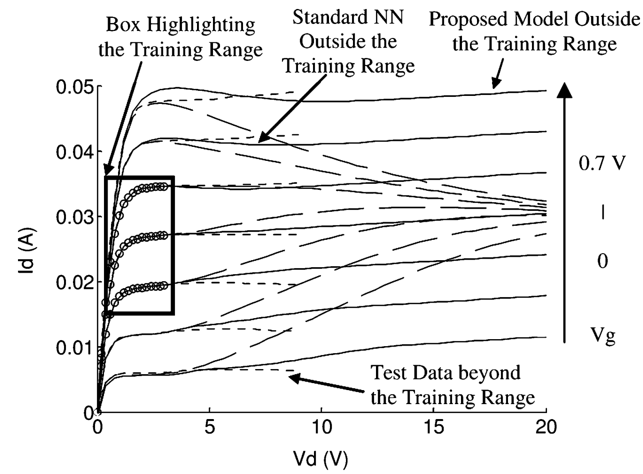

Fig. 2. Comparison of the drain current between the HEMT training data (o), the standard neural model (long dashed line), the proposed extrapolation (solid line), and the test data beyond training range (short dashed line). The training range is highlighted by the box. The proposed technique extrapolates far beyond the training range more meaningfully than the standard neural network.

III. Numerical Examples

A. Extrapolation on Neural Model of a Physics-Based HEMT

This example demonstrates the effect of the proposed neuron-level extrapolation by modeling the drain current behavior of a HEMT device from a physics-based device simulator *Synopsis Medici* [11]. The drain current of the HEMT is generated by solving the device Poisson equations in *Medici* at different gate and drain voltages. One set of training data is generated in the range of gate voltage $V_g = 0.2$–0.5 V, and drain voltage $V_d = 0.2$–3 V. Three sets of test data are generated in much wider ranges of V_g and V_d beyond training.

A neural network model is trained to learn the HEMT current-voltage behavior. Its extrapolation capability is tested by the test data sets. Fig. 2 compares the model behavior between the standard neural model and the proposed neuron-level extrapolation. It shows that when the inputs are far outside the training range, the drain current from the standard neural model converges to an arbitrary value because of saturation of neuron activation function. This is rectified by the proposed extrapolation, which maintains the correct tendency of the output current far beyond the training range.

For comparison purpose, existing model-level extrapolation techniques, i.e., based on the multi-dimensional training boundary of the input space ($\boldsymbol{x}$-space) [9], [10], are also applied to the same example. The 1st and 2nd order derivatives of the entire neural network function at the multi-dimensional training boundary are used for linear and quadratic extrapolations. Table I compares efficiency between the neural models with and without extrapolation. The proposed model is fast because its analytical formulation avoids the required derivative evaluation by the model-level extrapolation. Further speedup is because the original nonlinear switch function is replaced by the simple linear function in (4) for hidden neurons, which need to extrapolate. The results demonstrate that the proposed extrapolation is simple and effective with comparable accuracy and faster simulation.

TABLE I
COMPARISON OF ACCURACY AND SPEED BETWEEN DIFFERENT EXTRAPOLATION TECHNIQUES WITHIN AND BEYOND TRAINING FOR THE HEMT EXAMPLE

	Standard NN Without Extrapolation	NN with Proposed Neuron-Level Extrapolation	NN with Model-Level Linear Extrapolation	NN with Model-Level Quadratic Extrapolation
Training Error (Vg: 0.2-0.5V, Vd: 0.2-3V)	0.07%	0.07%	0.07%	0.07%
Test Error Beyond Training (Vg: 0.0-0.7V, Vd: 0-5V)	0.95%	0.71%	0.70%	0.71%
Test Error Beyond Training (Vg: 0.0-0.7V, Vd: 0-10V)	4.07%	1.62%	1.28%	1.96%
Test Error Beyond Training (Vg: 0.0-0.7V, Vd: 0-20V)	12.37%	3.57%	2.76%	2.74%
CPU for 1000 Simulations of the Extrapolation Test	5.52s	4.75s	7.30s	7.42s

TABLE II
COMPARISON OF ACCURACY AND SPEED BETWEEN DIFFERENT EXTRAPOLATION TECHNIQUES WITHIN AND BEYOND TRAINING POWER LEVEL AND FUNDAMENTAL FREQUENCY RANGE FOR THE POWER AMPLIFIER EXAMPLE

	Standard DNN Without Extrapolation	DNN with Proposed Neuron-Level Extrapolation	DNN with Model-Level Linear Extrapolation	DNN with Model-Level Quadratic Extrapolation
Training Error	0.46%	0.46%	0.46%	0.46%
Test Error 10% Beyond Training	1.15%	1.13%	1.19%	1.13%
Test Error 20% Beyond Training	8.05%	2.67%	3.49%	3.51%
Test Error 30% Beyond Training	17.51%	5.29%	7.76%	7.73%
CPU for 1000 Simulations of the Extrapolation Test	40.57s	38.97s	51.72s	53.81s

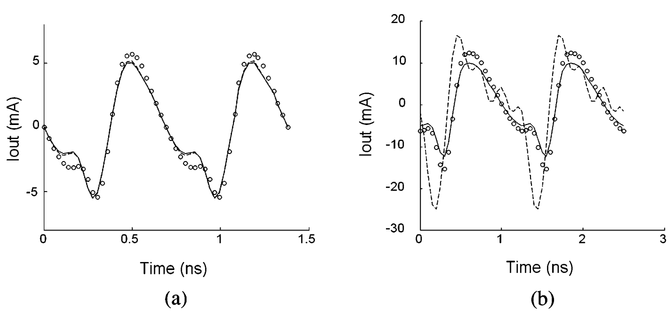

Fig. 3. Comparison of the output current of the power amplifier between the original circuit response (o), the standard DNN amplifier (– –), and the DNN amplifier with the proposed extrapolation (——). Harmonic balance simulations are performed in ADS with fundamental frequency and power level of (a) 10% and (b) 20% beyond the training range. The DNN with proposed extrapolation performs almost equally as the standard DNN for 10% extrapolation of the training range, and much better than the standard DNN when used far beyond the training range.

B. Extrapolation on Dynamic Neural Network Models of a Power Amplifier

This example addresses the application of the proposed neuron-level extrapolation to a power amplifier (PA) circuit simulation. The PA is modeled by two DNN behavioral models [6], one representing the nonlinear input impedance and the other representing the amplification and the output impedance. The training data for the DNNs are generated by harmonic balance (HB) simulations of the PA in ADS [12] using different input power levels (-10–0 dBm) and different fundamental frequencies (900–1400 MHz).

We examine the extrapolation by extending the input power level and fundamental frequency beyond that used in training. Fig. 3 compares the output current between solutions from original PA circuit, the standard DNN amplifier, and the DNN amplifier with proposed neuron-level extrapolation, showing that the DNN amplifier with the proposed extrapolation maintains better tendency than the standard DNN amplifier when used beyond training. The simplicity of the proposed extrapolation is also verified by comparing with the existing model-level extrapolations [9], [10]. Since the time-domain data required by DNN training are converted from frequency-domain HB data [6], the training boundary of the DNN input space (multi-dimensional $\boldsymbol{x}$-space) is irregular. This may lead to reduced accuracy and/or increased complexity (CPU time) in existing extrapolation techniques. As shown in Table II, with exact γ-boundary at the neuron level, the proposed extrapolation outperforms the existing technique with improved reliability and faster convergence in HB simulation.

REFERENCES

[1] Q. J. Zhang and K. C. Gupta, *Neural Networks for RF and Microwave Design*. Boston, MA: Artech House, 2000.
[2] Q. J. Zhang, K. C. Gupta, and V. K. Devabhaktuni, “Artificial neural networks for RF and microwave design: From theory to practice,” *IEEE Trans. Microw. Theory Tech.*, vol. 51, no. 4, pp. 1339–1350, Apr. 2003.
[3] P. Burrascano, S. Fiori, and M. Mongiardo, “A review of artificial neural networks applications in microwave computer-aided design,” *Int. J. RF Microw. CAE*, vol. 9, pp. 158–174, May 1999.
[4] V. Rizzoli, A. Neri, D. Masotti, and A. Lipparini, “A new family of neural network-based bi-directional and dispersive behavioral models for nonlinear RF/Microwave subsystems,” *Int. J. RF Microw. CAE*, vol. 12, pp. 51–70, Dec. 2002.
[5] J. E. Rayas-Sanchez, “EM-based optimization of microwave circuits using artificial neural networks: the state-of-the-art,” *IEEE Trans. Microw. Theory Tech.*, vol. 52, no. 1, pp. 420–435, Jan. 2004.
[6] J. J. Xu, M. C. E. Yagoub, R. T. Ding, and Q. J. Zhang, “Neural-based dynamic modeling of nonlinear microwave circuits,” *IEEE Trans. Microw. Theory Tech.*, vol. 50, no. 12, pp. 2769–2780, Dec. 2002.
[7] C. Brezinski and M. R. Zaglia, *Extrapolation Methods: Theory and Practice*. New York: North-Holland, 1991.
[8] T. Jan, “Robust short term prediction using combination of linear regression and modified probabilistic neural network model,” in *Proc. IEEE Int. Joint Conf. Neural Networks*, Portland, OR, Jul. 2003, pp. 2478–2481.
[9] L. C. Kiong, M. Rajeswari, and M. V. C. Rao, “Extrapolation detection and novelty-based node insertion for sequential growing multi-experts network,” *Appl. Soft Comput.*, vol. 3, no. 2, pp. 159–175, Sep. 2003.
[10] J. J. Xu, M. C. E. Yagoub, R. T. Ding, and Q. J. Zhang, “Robust neural based microwave modelling and design using advanced model extrapolation,” in *IEEE MTT-S Int. Dig.*, Fort Worth, TX, Jun. 2004, pp. 1549–1552.
[11] *Synopsys Medici*. Mountain View, CA: Synopsys, Inc., 2007.
[12] *Advanced Design System*. Palo Alto, CA: Agilent Technologies, 2006.